Simran Chauhan

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Education

Worcester Polytechnic Institute, Worcester, MA — Aug 2024 – May 2026

MS in Robotics Engineering — GPA: 3.8/4.0

Coursework: Computer Vision, Machine Learning, Robot Control, Robot Manipulation

Visvesvaraya National Institute of Technology, Nagpur, India — Jul 2017 – Jun 2021

BTech in Electrical Engineering — GPA: 8.59/10

Technical Skills

Languages	Python, C++
Perception & ML	PyTorch, OpenCV, Deep Learning, CNNs, Object Detection, Semantic Segmentation, Optical Flow, NeRF, Gaussian Splatting
Sensors & Fusion	LiDAR, Camera, IMU, Stereo Vision, Sensor Calibration, Kalman Filtering, MSCKF, Visual-Inertial Odometry
Robotics	ROS/ROS2, SLAM (ORB-SLAM, RTAB-SLAM), CARLA, Gazebo, RViz, Google Cartographer, LQR & MPC Control
Simulation & Tools	Unreal Engine, Cloud Compare, Blender, Synthetic Data Generation, Git, Linux, MATLAB

Research Experience

Graduate Research Assistant — Aug 2025 – May 2026

WPI Autonomous Vehicle Mobility Institute (AVMI)

Terrain Generation and Real-Time Heading-Aware Traversability Estimation for Off-Road AVs

- Built a multi-scale fractal terrain generator (Weierstrass–Mandelbrot layers) with fractal-dimension-driven difficulty control, exporting 16-bit heightmaps to Unreal Engine for large-scale off-road simulation.
- Generated ~8,400 heading-aware mobility labels for the MRZR using a deterministic radius-based topographical-mobility solver over 700 procedurally-generated terrains (50m×50m at 5cm/px).
- Trained a geometry-supervised CNN reproducing the solver in real time (~30 min → 10–20 ms/patch on GPU, 0.89 F1) and deployed it as a live ROS2 node in an Unreal Engine autonomy stack.

Simulation-Based Sensor Suite & Synthetic Data Pipeline for Off-Road Autonomy (Unreal Engine 5 + ROS2)

- Designed a synthetic-data engine in Unreal Engine 5 simulating LiDAR and camera sensors, streaming labeled point clouds, RGB, and semantic masks over ROS2 to remove manual annotation for off-road obstacle detection.
- Simulated spinning-LiDAR physics with configurable FOV/resolution, chunked 360° sweeps via parallel raycasting, and per-object classification with colored point-cloud output.
- Built a monocular camera sensor with real-time semantic segmentation via stencil-ID-to-color post-processing, producing pixel-aligned RGB/segmentation pairs for training.

Graduate Student Researcher — Jun 2025 – Nov 2025

Perception and Autonomous Robotics (PEAR) Lab, WPI

Controllable Depth-of-Field in Gaussian Splatting – VizFlyt UAV Simulator

- Implemented controllable depth-of-field rendering in a 3D Gaussian Splatting model (Splatfacto), with learnable focal-distance and aperture parameters jointly optimized alongside the scene geometry.
- Engineered a differentiable CUDA bokeh kernel (custom autograd.Function, hand-derived gradients) modeling physically accurate thin-lens defocus via depth-based circle-of-confusion, composited in linear gamma-corrected space.

Lensless Bio-Inspired Stereo Vision for Micro Aerial Vehicles

- Designed a single-pinhole lensless MLA camera that captures multiplexed overlapping views; formulated stereo reconstruction as a supervised demultiplexing problem and generated 20K synthetic training pairs using a distance-invariant homography.
- Built a two-pinhole servo-actuated imaging rig on an NVIDIA Jetson platform that selectively exposes two MLA lenslets to acquire aligned blended/left/right image triplets, enabling real-data supervision for stereo reconstruction.
- Trained a 50M-parameter dual-decoder U-Net with photometric and geometric losses (L1, SSIM, reconstruction, and soft epipolar consistency) to demultiplex blended captures into stereo views for depth estimation.

Projects

Deep and Un-Deep Visual-Inertial Odometry — Feb 2025 – Mar 2025

WPI

- Implemented a stereo MSCKF estimator on the error-state manifold with sliding-window pose augmentation and null-space-projected, QR-compressed EKF updates, reaching 0.094 m translational RMSE on EuRoC MH-01.
- Built a dual-stream learned VIO (FlowNet-style CNN + inertial LSTM) for 6-DoF pose on a custom Blender dataset, analyzing scale observability under loosely-coupled fusion.

Einstein Vision: Autonomous-Driving Perception Pipeline — Mar 2025 – Apr 2025
WPI

- Built a monocular-to-3D perception pipeline from single-camera dashcam footage, fusing YOLOv11x, Mask R-CNN, DETIC, DepthPro, and RAFT to detect and segment lanes, vehicles, pedestrians, signs, and traffic-light states; rendered reconstructed scenes in Blender.
- Recovered 3D lane geometry via ground-plane back-projection and Bézier curve fitting; built an OCR-based speed-sign reader and an HSV template-matching traffic-light classifier.
- Added temporal reasoning via optical-flow-based parked/moving classification and frame-to-frame 3D velocity estimation for time-to-contact collision prediction.

3D Reconstruction using Structure from Motion — Feb 2025 – Mar 2025
WPI

- Engineered an SfM pipeline integrating triangulation with RANSAC, minimizing reprojection errors by 89% and achieving 0.87-pixel reconstruction accuracy.
- Developed multi-view camera system with PnP algorithms and sparse bundle adjustment, improving pose estimation by 57% while optimizing 1,200+ 3D points simultaneously.

Camera Calibration (Zhang's Method) & Panorama Stitching — Jan 2025 – Mar 2025
WPI

- Implemented Zhang's calibration with Levenberg-Marquardt optimization, reducing reprojection error by 76% (2.15 → 0.511 pixels).
- Built panorama stitching with Harris corners + ANMS + RANSAC homography; developed dual deep learning approaches (supervised VGG-16 at ∼3px EPE; unsupervised photometric loss).

Additional Course Projects (links): AV Localization in CARLA (ORB-SLAM, RTAB-SLAM, ICP), Kalman Filter Sensor Fusion (87% noise reduction), PB-Lite Boundary Detection, CIFAR-10 Classification (ResNet/ResNeXt/DenseNet, 84%), Quadrotor LQR/PD Control, OpenManipulatorX Kinematics in ROS2.

Work Experience

Senior Technical Executive, Synergy Denmark A/S, India — Nov 2023 – Jul 2024
Led data integration for an internal AI tool (SYIA), liaising between end-users and the development team.

Technical Trainee (Electrical & Automation), Synergy Denmark A/S, India — Nov 2021 – Oct 2023
Built a Power BI tracking system for EE allocation, improving team workflow efficiency by 15%.

Technical Trainee, Maersk Tankers Pvt Ltd, India — Jul 2021 – Nov 2021
400+ hours of training in vessel electrical systems and wiring diagrams.

Publications

- **S. Chauhan**, et al. *Real-Time Heading-Aware Terrain Mobility Estimation for Off-Road Autonomous Vehicles.* In preparation, targeting IEEE Robotics and Automation Letters (RA-L), 2026.
- S. Zhang, **S. Chauhan**, et al. *Real-Time Mixed-Reality-Based Simulator for Autonomous Off-Road Vehicle Research.* Presented at the Intl. Conf. on Dynamic Games, Optimal Guidance and Applications in Autonomous Systems, Honolulu, HI, 2026.
- H. S. Parihar, . . . , **S. Chauhan**. *IoT-Based Ambiance Monitoring System.* ICAME, 2020.
- R. Runwal, . . . , **S. Chauhan**, N. Marne. *Hand Gesture Control of Computer Features.* ICAME, 2020.